Student Name: Ranjeet Kumar Sharma

Batch: June

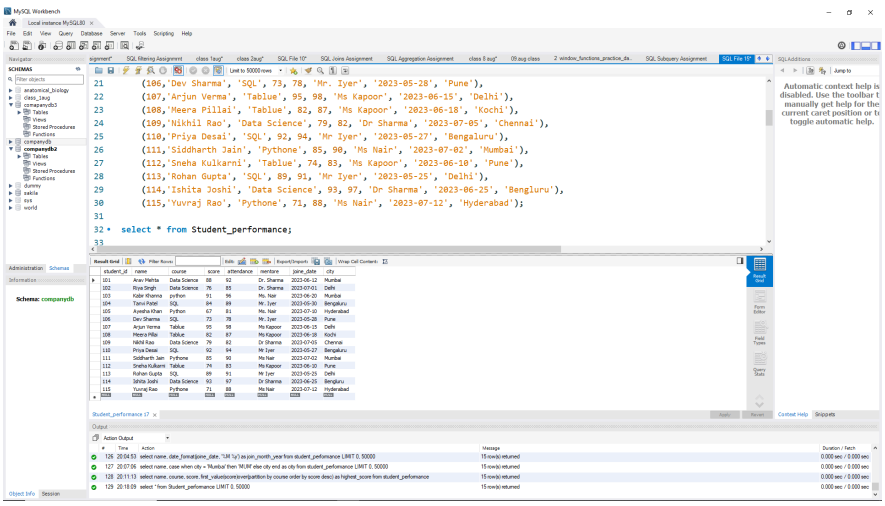
Course: Data Analytics with AI

Assignment: SQL Window Function Assignment

Submission Date:- 11-08-2026

Email_id: Ranjeet Kumar (Sharma)

➔ **Write the Question and Answers.**



Q 1 : Create a ranking of students based on score (highest first).

Ans:-select name, score,

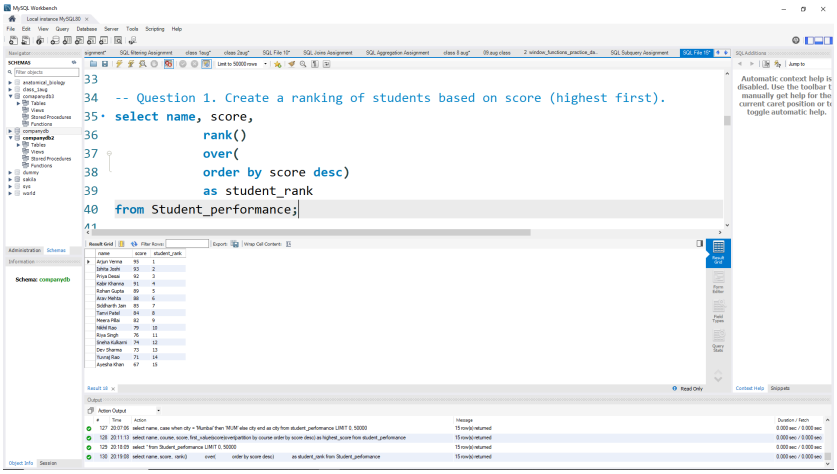
rank()

over(

order by score desc)

as student_rank

from Student_performance;



Q 2 : Show each student's score and the previous student's score (based on score order).

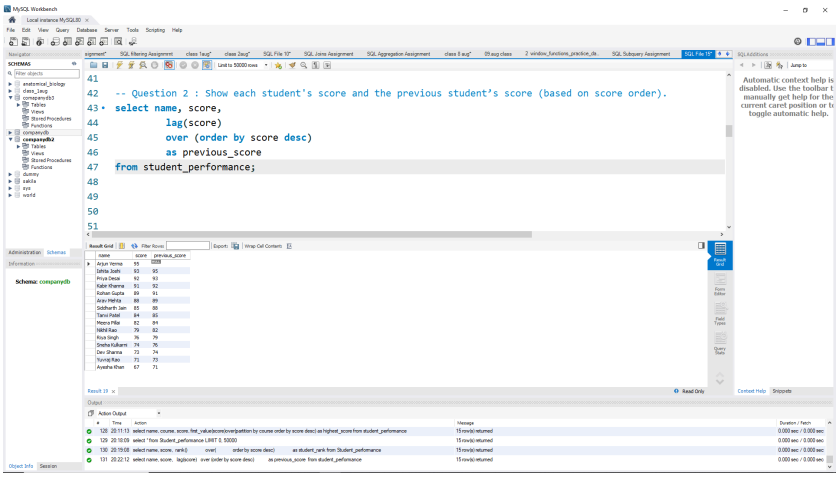
Ans:- select name, score,

lag(score)

over (order by score desc)

as previous_score

from student_performance;



Q 3 : Convert all student names to uppercase and extract the month name from join_date.

Ans:- select

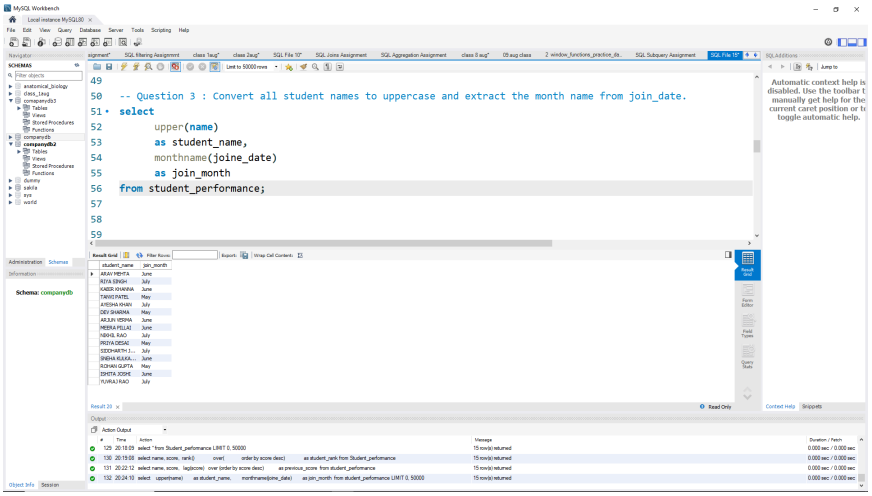
upper(name)

as student_name,

monthname(joine_date)

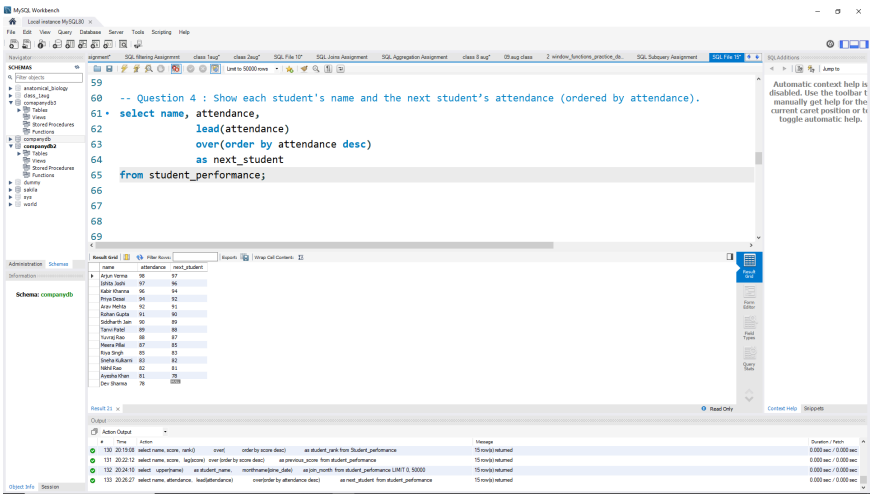
as join_month

from student_performance;



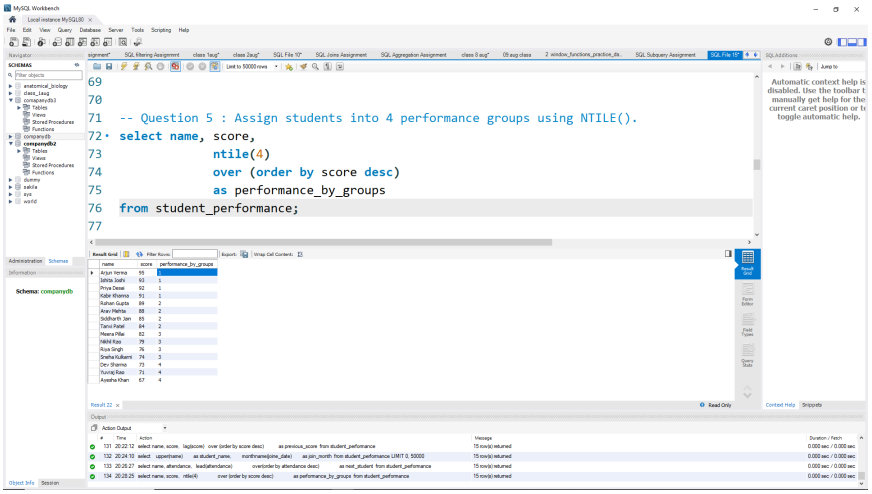
Q 4 : Show each student's name and the next student’s attendance (ordered by attendance).

Ans:- select name, attendance,
lead(attendance)
over(order by attendance desc)
as next_student
from student_performance;



Q 5 : Assign students into 4 performance groups using NTILE().

Ans:- select name, score,
ntile(4)
over (order by score desc)
as performance_by_groups
from student_performance;



Q 6 : For each course, assign a row number based on attendance (highest first).

Ans:-select name, course, attendance,

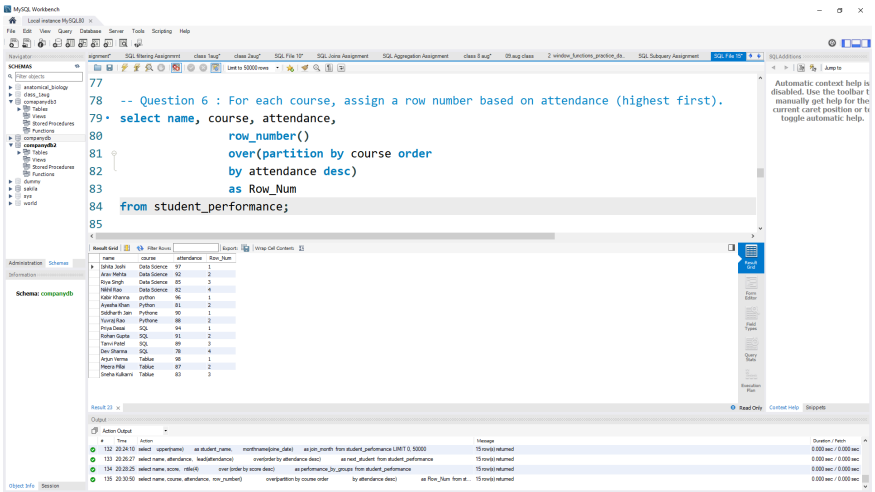
row_number()

over(partition by course order

by attendance desc)

as Row_Num

from student_performance;



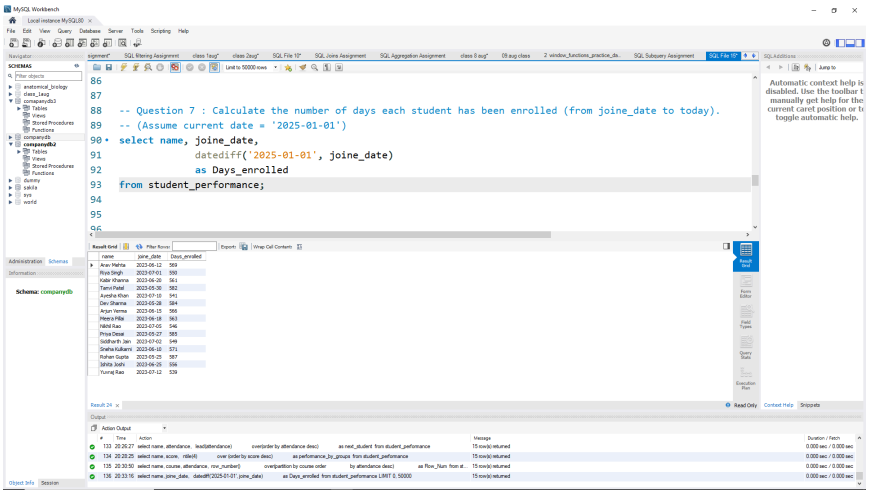
Q 7 : Calculate the number of days each student has been enrolled

(from join_date to today). (Assume current date = '2025-01-01')

Ans:- select name, joine_date,

datediff('2025-01-01', joine_date)

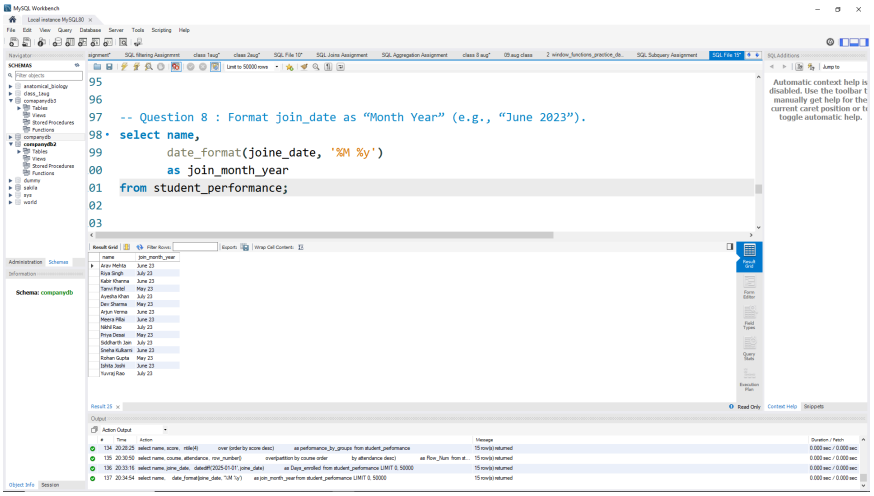
```
        as Days_enrolled
    from student_performance;
```



Q 8 : Format join_date as “Month Year” (e.g., “June 2023”).

Ans:-

```
select name,
        date_format(joine_date, '%M %y')
        as join_month_year
    from student_performance;
```

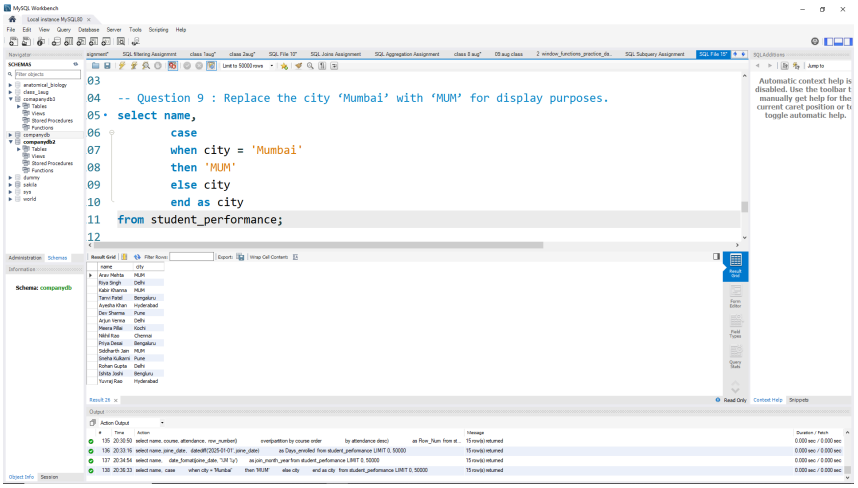


Q 9 : Replace the city ‘Mumbai’ with ‘MUM’ for display purposes.

Ans:-

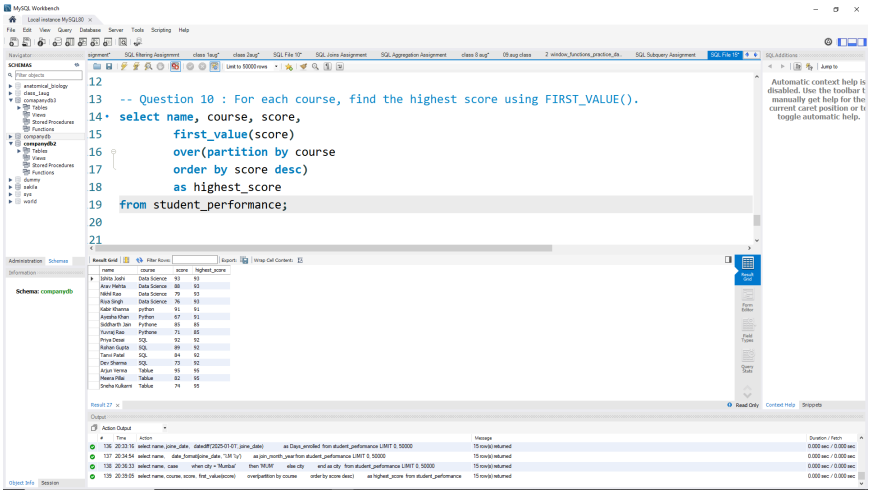
```
select name,
        case
        when city = 'Mumbai'
    then 'MUM'
```

```
        else city
        end as city
from student_performance;
```



Q 10 : For each course, find the highest score using FIRST_VALUE()

Ans:-
```
select name, course, score,
        first_value(score)
        over(partition by course
        order by score desc)
        as highest_score
from student_performance;
```



Conclusion:

In this assignment, I practiced SQL Window Functions and various SQL functions such as `RANK()`, `LAG()`, `LEAD()`, `NTILE()`, `ROW_NUMBER()`, `DATEDIFF()`, `DATE_FORMAT()`, and `FIRST_VALUE()`. These functions helped me understand how to analyze, rank, compare, and transform data efficiently using SQL.

Submitted by: Ranjeet Sharma

Topic: SQL Window Functions

Tool: MySQL Workbench